



Pressure loss in turbulent flow

In flow of this nature, the pressure loss is associated with the shear stresses caused by the frictional resistance of the airway wall.

15

Tuesday
July 2008

The loss of pressure in circular airways may be estimated by Darcy's equation

$$\Delta P = \frac{4f \rho L \bar{V}^2}{2D}$$

where ΔP = pressure loss in Pa

f = an experimentally determined friction factor (dimensionless) or Darcy's coefficient of resistance.

ρ = air density (kg/m^3)

L = airway length (m)

$\bar{V}$ = mean air velocity (m/s) and

D = airway diameter (m)

It is clear from the above eqn. that the pressure loss is proportional to the square of the mean velocity $\bar{V}^2$.

For non-circular airways, the above eqn. can be used if the diameter D is replaced by the equivalent diameter, defined as

$$D_e = \frac{4A}{P}$$

MEMO / TO DO where

D_e = equivalent diameter (m)

A = cross-sectional area of the airway (m^2), and

P = perimeter of the airways in contact with air flowing (m)

AUGUST 2008

S	M	T	W	T	F	S
31					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

16

Wednesday

July 2008

The above eqⁿ. then becomes

$$P = \frac{4fPLV^2\rho}{2 \times 4A}$$



STATE OF AIR FLOW IN MINE OPENINGS

Reynolds Number : In fluid dynamics three states of fluid flow are observed: 1. Laminar 2. intermediate and 3. Turbulent

The nature of flow of a fluid (laminar or turbulent) is governed by a dimensionless parameter called the Reynolds number.

The Reynolds no. is defined as

$$Re = \frac{\rho V D}{\mu} \quad \text{or} \quad \frac{V D}{\nu}$$

where Re = The Reynolds no. (dimensionless)

V = mean air velocity (m/s)

ρ = air density (kg/m³)

D = airway diameter (m)

μ = dynamic viscosity of air (Pa.s) and

ν = kinematic viscosity of air (m²/s)

The Reynolds no. determines the relative influence of viscosity on the flow pattern. The larger the Reynolds no. the lesser is the influence of viscosity on the flow pattern and at high Reynolds nos. the viscous resistance to deformation becomes less important in comparison to inertial resistance to acceleration, which is a function of the fluid density.

JULY 2008						
S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

MEMO / TO DO

The value of the Reynolds no. determines whether laminar or turbulent flow prevails.



If $Re \leq 2000$ the flow is laminar and

if $Re \geq 4000$ " " turbulent

Between these values is a critical zone in which the flow is transitional.

17

Thursday
July 2008

As Reynolds no. increases beyond a certain critical limit, laminar flow becomes unstable.

Problem:

Determine the nature of flow in a mine airway of 2.2×1.8 m size if the velocity of air in the airway is 0.5 m/s . Calculate the max. velocity at which the flow still remains truly laminar. Assume the air to be at an atmospheric press. of 101.33 kPa and at a temp. of 303 K .

Other data: at 303 K , the kinematic viscosity $\nu = 0.16 \times 10^{-4} \text{ m}^2/\text{s}$ of air

Soln.

$$Re = \frac{V D_e}{\nu}$$

where V = air velocity 0.5 m/s

$$\nu = 0.16 \times 10^{-4} \text{ m}^2/\text{s}$$

$$D_e = \text{equivalent dia. of the airway} = \frac{4A}{P} = \frac{4 \times 2.2 \times 1.8}{2(2.2 + 1.8)} = 1.98 \text{ m.}$$

$$\text{Therefore, } Re = \frac{0.5 \times 1.98}{0.16 \times 10^{-4}} = 6187.5$$

Since $Re > 2000$, the flow is turbulent.

MEMO / TO DO

For the flow to be laminar $Re \leq 2000$

$$\Rightarrow 2000 = \frac{V \cdot D_e}{\nu} \Rightarrow V = \frac{2000 \times \nu}{D_e} = \frac{2000 \times 0.16 \times 10^{-4}}{1.98} = 0.16 \text{ m/s}$$

AUGUST 2008

S M T W T F S

31 1 2

3 4 5 6 7 8 9

10 11 12 13 14 15 16

17 18 19 20 21 22 23

24 25 26 27 28 29 30

18

Friday

July 2008

Relationship between Darcy's friction factor and the Reynolds number



It is convenient to compare the pressure loss given by Hagen - Poiseuille's and Darcy's equations, i.e.

$$\Delta P = \frac{32\mu LV}{D^2} = \frac{4fLV^2}{2D}$$

Therefore,

$$f = \frac{16\mu}{\rho V D} = \frac{16}{Re}$$

In turbulent flow, 'f' is a function of the Reynolds no. and of the relative roughness of the airway.

The relative roughness is the ratio of the average height of the surface irregularities (ϵ) to the airway diameter (D)

$$\therefore \text{Relative roughness} = \frac{\text{average ht. of the surface irregularities } (\epsilon)}{\text{Airway diameter } (D)}$$

Moody's Diagram:

The relationship between f , Re and ϵ/D is given in Moody's Diagram which enables to calculate the pressure loss for shafts, raises and lined roadways.

JULY 2008						
S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

MEMO / TO DO



Resistance coefficients / Friction factors

The resistance coefficient ' f ' is a function of the Reynolds number.

19

Saturday

July 2008

For values of $Re < 2000$ (laminar flow)

$$f = \frac{64}{Re}$$

This relationship follows from the equation of Poiseuille:

$$\begin{aligned} \Delta P &= \frac{32 \mu \bar{V} L}{D^2} = \frac{64 \mu}{D} \cdot \frac{L}{D} \cdot \frac{\bar{V}^2}{2} \\ &= \frac{64}{(\bar{V} D / \mu)} \cdot \frac{L}{D} \cdot \frac{\bar{V}^2}{2} = \frac{64}{Re} \cdot \frac{L}{D} \cdot \frac{\bar{V}^2}{2} \\ &= f \cdot \frac{L}{D} \cdot \frac{\bar{V}^2}{2} \end{aligned}$$

For values of $Re > 2000$ (turbulent flow)

July 20, Sunday

$$f = \frac{0.316}{Re^{0.25}} \text{ is given by Blasius}$$

Above eqn. holds true to a value of $Re = 100,000$, beyond which ' f ' is given by the Karman-Prandtl resistance equation for turbulent flow in smooth pipes:

$$\frac{1}{\sqrt{f}} = 2 \log (Re \sqrt{f}) - 0.8$$

MEMO / TO DO

AUGUST 2008

S	M	T	W	T	F	S
31					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

21

Monday

July 2008

Airflow through mine openings

Pressure loss and resistance coefficient for mine airways (Atkinson's equation)



Mine airways can be considered as very rough conduits with incompressible flow.

In mine ventilation, an alternative friction factor K is proposed by Atkinson instead of f .

Atkinson arrived at this friction factor by assuming the pressure loss to be proportional to

- the airway length
- the perimeter
- the square of the air velocity and
- inversely proportional to the cross-sectional area

Thus,

$$\Delta P = \frac{K \cdot L \cdot P \cdot V^2}{A}$$

$$= \frac{K \cdot S \cdot V^2}{A} = \frac{K \cdot S \left(\frac{Q}{A}\right)^2}{A} = \frac{K S Q^2}{A^3}, \text{ Pa}$$

(For a given airway $R = \frac{K S}{A^3}$ is constant)

$$\Delta P = R Q^2$$

Where ΔP = pressure loss in Pa,
 K = friction factor or coefficient of friction (norm lies between 0.1-0.2) (Ns^2/m^4 or kg/m^3)

L = airway length (m), R = Resistance of airway (Ns^2/m^8)

JULY 2008						
S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

MEMO / TO DO

P = airway perimeter (m)

V = airflow velocity (m/s)

A = airway cross sectional area (m^2)

$S = L \cdot P$ = area of rubbing surface (m^2)

Q = volume flow rate (m^3/s)

$$= \frac{P}{Q^2}$$



Atkinson ignored the density effects in deriving the above equation and it is valid only for a standard air density of 1.2 kg/m^3

22

Tuesday

July 2008

The form of equation used at any density of air is

$$\Delta P = \frac{K \cdot L \cdot P \cdot V^2}{A} \times \frac{1}{1.2} = \frac{K \cdot L \cdot P \cdot Q^2}{A^3} \times \frac{1}{1.2}$$

Equating

$$\frac{KLPV^2}{A} \cdot \frac{1}{1.2} = \frac{4f \cdot L \cdot P \cdot V^2}{2 \times 4A}$$

Derived from Darcy's eqn.

$$\Delta P = \frac{4f \cdot L \cdot V^2}{2D}$$

for non-circular airway

$$D = D_e = \frac{4A}{P}$$

$$\Rightarrow \Delta P = \frac{4f \cdot L \cdot P \cdot V^2}{2 \times 4A}$$

We have

$$\frac{1}{2} = \frac{K}{1.2}$$

or

$$K = 0.6f$$

for a given airway, $\frac{KLP}{A^3}$ is constant and is substituted

by the term R , the resistance of the airway. In that case,

$$\Delta P = R \cdot Q^2 \times \frac{1}{1.2}$$

The unit of R is Ns^2/m^8

Value of friction factor, K

It normally lies between 0.01 and 0.02 Ns^2/m^4 .

Wherever, values obtained from pressure-volume flow rate surveys should be used.

MEMO / TO DO

AUGUST 2008

S M T W T F S

31 1 2

3 4 5 6 7 8 9

10 11 12 13 14 15 16

17 18 19 20 21 22 23

24 25 26 27 28 29 30

AUG

SEP

OCT

NOV

DEC